Customer Churn Analysis

Analytics report | Synthetic customer dataset

Objective: calculate baseline churn, assess customer value and monthly revenue at risk, identify behavioral patterns, and recommend retention actions.

1. Data preparation

Original dataset: 605 rows and 13 columns. Removed 5 exact duplicate rows. Filled 13 missing Region values with “Unknown”. Converted TotalChargesINR to numeric; 10 unknown values were imputed as MonthlyChargesINR × TenureMonths. This is an estimate, not a verified billing amount.

2. Baseline metrics

Metric	Result
Rows after cleaning	600
Churned customers	247
Baseline churn rate	41.17%
Retained customers	353
Estimated monthly recurring revenue loss (churned customers)	■240,102.48
Average monthly charge	■869.03
Mean historical value proxy among churned customers	■24,440.33

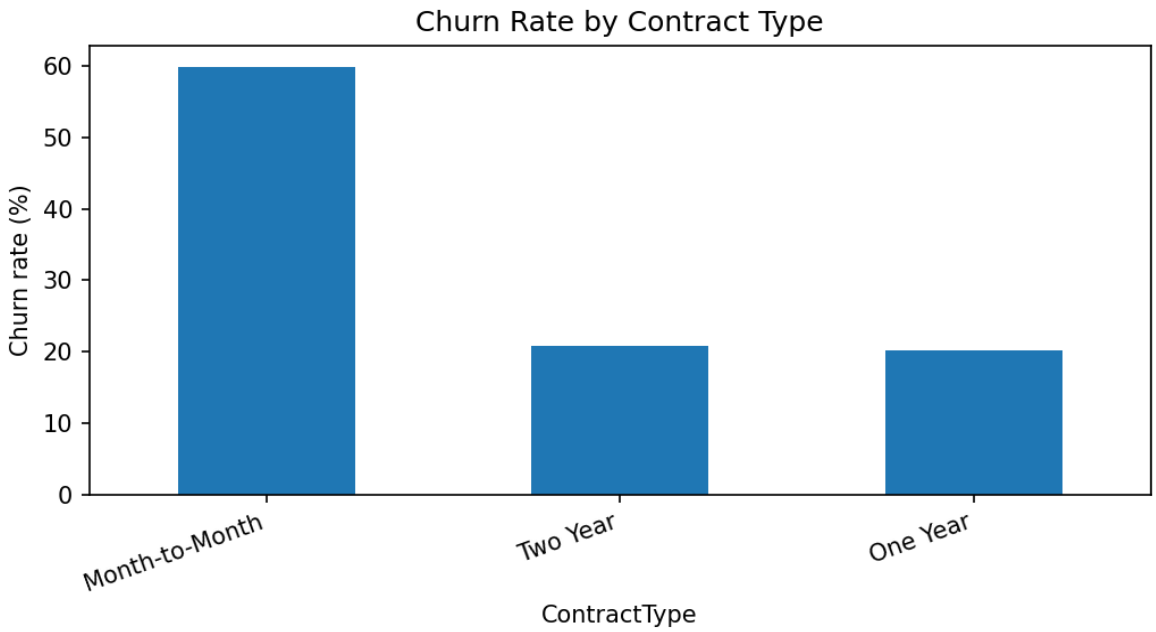
CLV note: the dataset has no expected future lifespan, margin, or discount-rate fields. Therefore, CustomerLifetimeValueINR is reported as observed historical TotalChargesINR (a lifetime-value proxy), not a predictive future CLV.

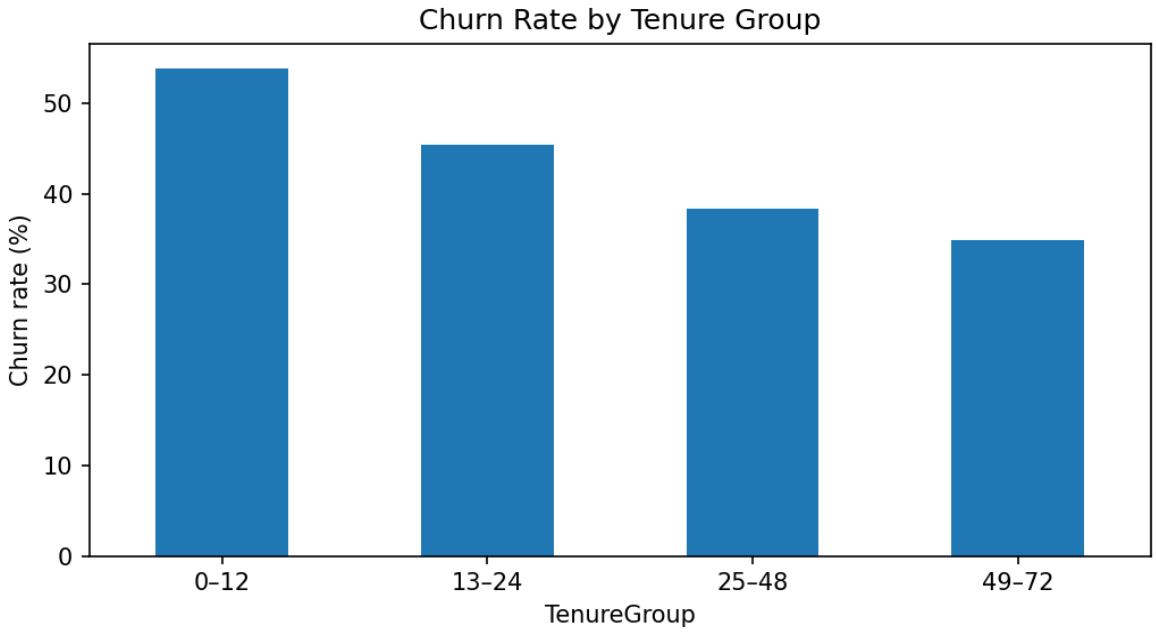
3. Churn patterns

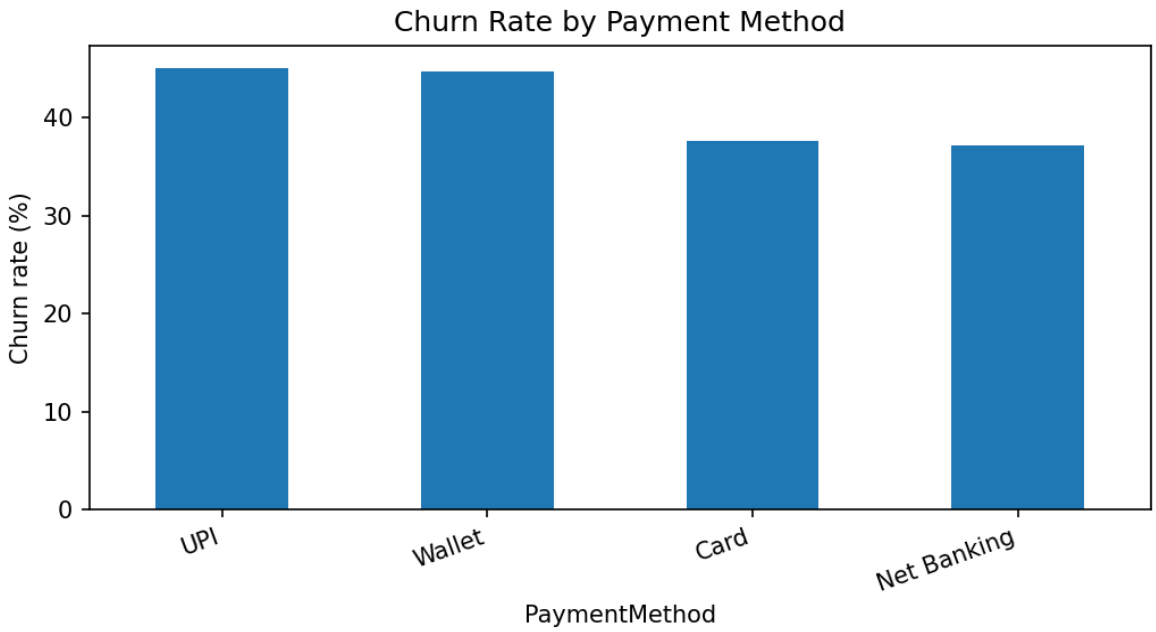
Contract-type churn rates: Month-to-Month: 59.8%; Two Year: 20.8%; One Year: 20.1%.

Tenure-group churn rates: 0–12: 53.8%; 13–24: 45.5%; 25–48: 38.3%; 49–72: 34.8%.

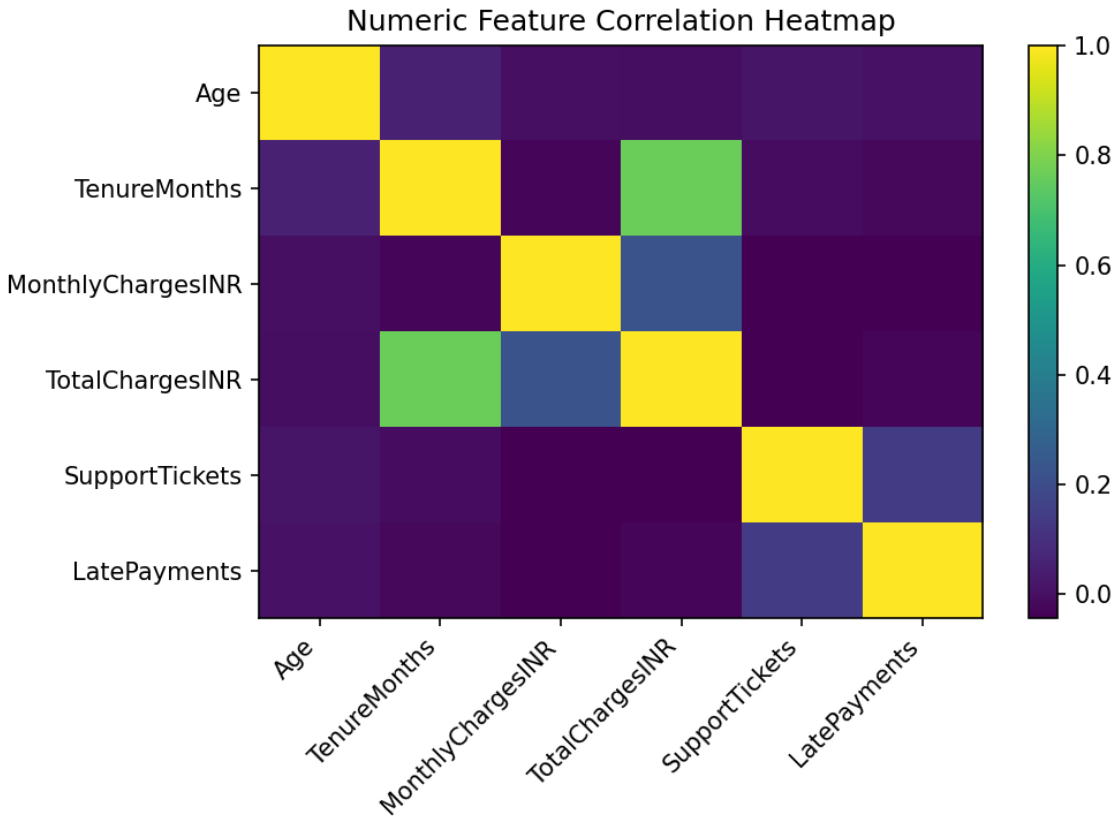
Payment-method churn rates: UPI: 45.1%; Wallet: 44.7%; Card: 37.6%; Net Banking: 37.1%.


Churn Rate by Contract Type
Churn rate (%)
0
10
20
30
40
50
60
Month-to-Month
Two Year
One Year
ContractType


Churn Rate by Tenure Group
Churn rate (%)
0
10
20
30
40
50
0–12
13–24
25–48
49–72
TenureGroup


Churn Rate by Payment Method
Churn rate (%)
0
10
20
30
40
UPI
Wallet
Card
Net Banking
PaymentMethod

4. Numeric relationship heatmap



The heatmap shows pairwise correlations among numeric variables. Correlation is descriptive and does not establish causation.

5. Business recommendations

1. Prioritize retention outreach for month-to-month customers; test annual-plan incentives and explain the value of longer contracts.

2. Create an early-tenure onboarding journey (first 30–90 days) with check-ins, setup help, and usage tips.

3. Contact customers with repeated support tickets or late payments before renewal; resolve recurring issues and offer proactive billing reminders.

4. Offer easy-to-use autopay/payment options and investigate payment friction by method.

5. Track churn and revenue-at-risk monthly by contract, tenure, payment method, and support activity; compare intervention cohorts against a baseline.

6. Limitations

The records are synthetic. Some TotalCharges values were imputed. The dataset is cross-sectional and lacks dates, gross margin, acquisition cost, and forecast horizon; consequently MRR loss is a snapshot of monthly charges associated with churned records, and predictive CLV cannot be estimated robustly.